

Home
└─(kali㉿kali)-[~]
\$ mkdir -p ~/cip-b101-lab1a

File System
└─(kali㉿kali)-[~]
\$ cd ~/cip-b101-lab1a

Tool
└─(kali㉿kali)-[~/cip-b101-lab1a]
\$ pwd
/home/kali/cip-b101-lab1a

└─(kali㉿kali)-[~/cip-b101-lab1a]
\$ echo \$((2#101010))

42

└─(kali㉿kali)-[~/cip-b101-lab1a]
\$ echo \$((16#AB2))
2738

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ echo "obase=2;126" | bc
1111110
```

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ printf "%X"
dquote>
dquote>
```

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ python3 - <<'PY'
print(int('101010', 2))
print(int('AB2', 16))
print(bin(126))
print(hex(2738))
PY

42
2738
0b1111110
0xab2
```

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ echo -n "Hello" > evidence_text.txt
cat evidence_text.txt
xxd evidence_text.txt
xxd -p evidence_text.txt
Hello00000000: 4865 6c6c 6f
48656c6c6f
```

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ printf "Hello"
Hello
```

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ printf "48656c6c6f" | xxd -r -p
Hello
```

Hello

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ python3 - <<'PY'
import binascii
text = b'Hello'
print('ASCII bytes:', text)
print('ASCII to hex:', binascii.hexlify(text).decode())
print('Hex to ASCII:', binascii.unhexlify('48656c6c6f').decode())
print('ord(H):', ord('H'))
print('chr(72):', chr(72))
PY
ASCII bytes: b'Hello'
ASCII to hex: 48656c6c6f
Hex to ASCII: Hello
ord(H): 72
chr(72): H

(kali㉿kali)-[~/cip-b101-lab1a]
$ echo "Digital Forensics Begins With Bytes." > Fuseini_lab1a.txt
```

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ cat Fuseini_lab1a.txt
Digital Forensics Begins With Bytes.

(kali㉿kali)-[~/cip-b101-lab1a]
$ xxd Fuseini_lab1a.txt
00000000: 4469 6769 7461 6c20 466f 7265 6e73 6963  Digital Forensic
00000010: 7320 4265 6769 6e73 2057 6974 6820 4279  s Begins With By
00000020: 7465 732e 0a                                tes..

(kali㉿kali)-[~/cip-b101-lab1a]
$ xxd -p Fuseini_lab1a.txt
4469676974616c20466f72656e7369637320426567696e73205769746820
42797465732e0a

(kali㉿kali)-[~/cip-b101-lab1a]
$ xxd -r -p <(xxd -p Fuseini_lab1a.txt)
Digital Forensics Begins With Bytes.
```

```
(kali㉿kali)-[~/cip-b101-lab1a]  
$ date +%s  
1782253076
```

```
(kali㉿kali)-[~/cip-b101-lab1a]  
$ date -u -d @1643643491  
Mon Jan 31 03:38:11 PM UTC 2022
```

```
(kali㉿kali)-[~/cip-b101-lab1a]  
$ python3 - <<'PY'  
from datetime import datetime, timezone  
print(datetime.fromtimestamp(1643643491, tz=timezone.utc))  
PY  
2022-01-31 15:38:11+00:00
```

```
(kali㉿kali)-[~/cip-b101-lab1a]  
$ date  
date -u  
python3 - <<'PY'  
from datetime import datetime, timezone  
print('Local:', datetime.now())  
print('UTC  :', datetime.now(timezone.utc))  
PY  
Tue Jun 23 06:19:24 PM EDT 2026  
Tue Jun 23 10:19:24 PM UTC 2026  
Local: 2026-06-23 18:19:24.956634  
UTC   : 2026-06-23 22:19:24.956854+00:00
```

```
(kali㉿kali)-[~/cip-b101-lab1a]
$ python3 - <<'PY'
from datetime import datetime, timezone
unix_epoch = datetime(1970, 1, 1, tzinfo=timezone.utc)
mac_epoch = datetime(2001, 1, 1, tzinfo=timezone.utc)
print(int((mac_epoch - unix_epoch).total_seconds()))
mac_value = 725846400
unix_value = mac_value + 978307200
print(datetime.fromtimestamp(unix_value, tz=timezone.utc))
PY

978307200
2024-01-02 00:00:00+00:00

(kali㉿kali)-[~/cip-b101-lab1a]
$ echo "Hello" > hello_with_newline.txt
echo -n "Hello" > hello_without_newline.txt

xxd hello_with_newline.txt
xxd hello_without_newline.txt
```

```
md5sum hello_with_newline.txt hello_without_newline.txt
sha256sum hello_with_newline.txt hello_without_newline.txt

00000000: 4865 6c6c 6f0a                Hello.
00000000: 4865 6c6c 6f                Hello
09f7e02f1290be211da707a266f153b3  hello_with_newline.txt
8b1a9953c4611296a827abf8c47804d7  hello_without_newline.txt
66a045b452102c59d840ec097d59d9467e13a3f34f6494e539ffd32c1bb35f18  hello_with_newline.txt
185f8db32271fe25f561a6fc938b2e264306ec304eda518007d1764826381969  hello_without_newline.txt

(kali㉿kali)-[~/cip-b101-lab1a]
$ python3 - <<'PY'
import sys
print('System byte order:', sys.byteorder)
PY

System byte order: little
```